

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Database Management Systems
COURSE CODE: CSA05

COURSE OUTCOME COVERED

CO3: *Analyze relational databases using functional dependencies, normalization (1NF to 5NF), and key constraints to identify redundancy and ensure data integrity. (BL2, BL3)*

Activity Tool : MCQ Test (Low)
Assessment Tool Weightage: 30%

QUESTIONS			Total Marks: 50
Q.No	Question	Bloom's Level	Marks
1	Which of the following is true about 1NF? (a) Allows multi-valued attributes (b) Eliminates partial dependencies (c) All attributes must be atomic (d) Eliminates transitive dependencies	BL2 – Understand	5
2	A relation is in 2NF if it is in 1NF and: (a) Has no transitive dependencies (b) Has no partial dependencies on primary key (c) Every attribute is a prime attribute (d) Has no multi-valued dependencies	BL2 – Understand	5
3	Which normal form deals with multi-valued dependencies? (a) 2NF (b) 3NF (c) BCNF (d) 4NF	BL2 – Understand	5
4	BCNF is a stricter version of: (a) 1NF (b) 2NF (c) 3NF (d) 4NF	BL2 – Understand	5
5	Functional dependency $X \rightarrow Y$ means: (a) Y determines X (b) X determines Y (c) X and Y are unrelated (d) Both determine each other	BL2 – Understand	5
6	Which of the following anomaly is NOT associated with unnormalized relations? (a) Insertion anomaly (b) Deletion anomaly (c) Update anomaly (d) Retrieval anomaly	BL2 – Understand	5
7	A superkey is: (a) A minimal set of attributes that uniquely identifies tuples (b) Any set of attributes that uniquely identifies tuples (c) An	BL2 – Understand	5

	attribute that depends on a non-prime attribute (d) A foreign key reference		
8	If R(A,B,C) has FD: $A \rightarrow B$ and $B \rightarrow C$, then C is: (a) Partially dependent on A (b) Fully dependent on A (c) Transitively dependent on A (d) Independently determined	BL2 – Understand	5
9	5NF is also known as: (a) Domain-Key Normal Form (b) Boyce-Codd Normal Form (c) Project-Join Normal Form (d) Element Normal Form	BL2 – Understand	5
10	Lossless-join decomposition ensures: (a) No data is lost during decomposition (b) All FDs are preserved (c) The relation is in BCNF (d) No null values exist	BL2 – Understand	5

Code of Conduct:

I KARNAM HARSHITH - 192511384 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: RUBRICS FOR CALCULATION:

MCQ					
Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Conceptual Understanding	30%	Demonstrates thorough understanding; answers all questions accurately	Shows good understanding with few errors	Partial understanding; several incorrect responses	Lacks understanding; majority of answers incorrect
Accuracy of Answers	25%	All answers are correct	Most answers are correct with minor mistakes	Some correct answers; frequent errors	Mostly incorrect answers
Application of Knowledge	20%	Correctly applies concepts to problem-based questions	Applies concepts with minor errors	Limited ability to apply concepts	Unable to apply concepts
Time Management	15%	Completes quiz well within time efficiently	Completes on time with minor delays	Requires extra time or rushes through questions	Unable to complete within given time
Question Interpretation	10%	Interprets all questions correctly and responds appropriately	Misinterprets a few questions	Often misinterprets questions	Unable to understand questions

1. Which of the following is true about 1NF?

- (a) Allows multi-valued attributes
- (b) Eliminates partial dependencies
- (c) All attributes must be atomic ✓**
- (d) Eliminates transitive dependencies

Correct Option: (c) All attributes must be atomic

Justification: First Normal Form (1NF) requires every attribute to contain only a single, indivisible value. A column should not contain multiple values or repeating groups in one cell. For example, storing DBMS, Python in a single Courses cell violates 1NF. Partial dependencies are handled by 2NF, while transitive dependencies are handled by 3NF. Therefore, the defining requirement of 1NF is atomic attribute values.

2. A relation is in 2NF if it is in 1NF and:

- (a) Has no transitive dependencies
- (b) Has no partial dependencies on primary key ✓**
- (c) Every attribute is a prime attribute
- (d) Has no multi-valued dependencies

Correct Option: (b) Has no partial dependencies on primary key

Justification: A relation must first satisfy 1NF before it can be in 2NF. 2NF specifically removes partial dependencies, where a non-key attribute depends on only part of a composite candidate key. For example, if (StudentID, CourseID) is the key and StudentName depends only on StudentID, it is a partial dependency. Transitive dependencies are addressed by 3NF. Multi-valued dependencies are addressed by 4NF.

3. Which normal form deals with multi-valued dependencies?

- (a) 2NF
- (b) 3NF
- (c) BCNF
- (d) 4NF ✓**

Correct Option: (d) 4NF

Justification: Fourth Normal Form (4NF) specifically deals with multi-valued dependencies (MVDs). A multi-valued dependency occurs when one attribute independently determines multiple values of another attribute. For example, a teacher may have multiple subjects and multiple hobbies independently. 4NF separates these

independent multi-valued relationships into different relations. Thus, 4NF reduces redundancy caused by multi-valued dependencies.

4. BCNF is a stricter version of:

- (a) 1NF
- (b) 2NF
- (c) 3NF ✓**
- (d) 4NF

Correct Option: (c) 3NF

Justification: Boyce-Codd Normal Form (BCNF) is a stronger form of Third Normal Form (3NF). In BCNF, for every non-trivial functional dependency $X \rightarrow Y$, X must be a superkey. 3NF has slightly less strict conditions and can allow some dependencies that BCNF does not. Therefore, every BCNF relation satisfies 3NF, but not every 3NF relation necessarily satisfies BCNF.

5. Functional dependency $X \rightarrow Y$ means:

- (a) Y determines X
- (b) X determines Y ✓**
- (c) X and Y are unrelated
- (d) Both determine each other

Correct Option: (b) X determines Y

Justification: The functional dependency $X \rightarrow Y$ means that the value of X uniquely determines the value of Y . If two tuples have the same value of X , they must have the same value of Y . For example, $\text{StudentID} \rightarrow \text{StudentName}$ means one StudentID identifies one StudentName . It does not automatically mean that Y determines X . Therefore, X is called the determinant and Y is the dependent attribute.

6. Which of the following anomaly is NOT associated with unnormalized relations?

- (a) Insertion anomaly
- (b) Deletion anomaly
- (c) Update anomaly
- (d) Retrieval anomaly ✓**

Correct Option: (d) Retrieval anomaly

Justification: The three standard anomalies associated with poorly designed or unnormalized relations are insertion, deletion, and update anomalies. An insertion

anomaly occurs when new information cannot be inserted without unrelated information. A deletion anomaly occurs when deleting one record unintentionally removes other useful information. An update anomaly occurs when the same information must be updated in multiple rows. Retrieval anomaly is not one of the standard normalization anomalies.

7. A superkey is:

- (a) A minimal set of attributes that uniquely identifies tuples
- (b) Any set of attributes that uniquely identifies tuples ✓**
- (c) An attribute that depends on a non-prime attribute
- (d) A foreign key reference

Correct Option: (b) Any set of attributes that uniquely identifies tuples

Justification: A superkey is any set of one or more attributes that can uniquely identify every tuple in a relation. It does not have to be minimal. For example, if StudentID uniquely identifies a student, then {StudentID} is a superkey, and {StudentID, Name} is also a superkey. A candidate key is different because it is a minimal superkey. Therefore, option (b) is correct.

8. If $R(A,B,C)$ has FD: $A \rightarrow B$ and $B \rightarrow C$, then C is:

- (a) Partially dependent on A
- (b) Fully dependent on A
- (c) Transitively dependent on A ✓**
- (d) Independently determined

Correct Option: (c) Transitively dependent on A

Justification: The given dependencies are $A \rightarrow B$ and $B \rightarrow C$. Therefore, A determines C indirectly through B: $A \rightarrow B \rightarrow C$. This is called a transitive dependency. C does not directly depend on A in the given dependency set; it is determined through the intermediate attribute B. Transitive dependencies are important when converting relations to 3NF. Hence, option (c) is correct.

9. 5NF is also known as:

- (a) Domain-Key Normal Form
- (b) Boyce-Codd Normal Form
- (c) Project-Join Normal Form ✓**
- (d) Element Normal Form

Correct Option: (c) Project-Join Normal Form

Justification: Fifth Normal Form (5NF) is also called Project-Join Normal Form (PJ/NF). It deals primarily with join dependencies and ensures that a relation cannot be further decomposed into smaller relations without losing the ability to reconstruct the original relation correctly. It is used to eliminate redundancy caused by complex join relationships. Therefore, 5NF is correctly identified as Project-Join Normal Form..

10. Lossless-join decomposition ensures:

(a) No data is lost during decomposition ✓

(b) All FDs are preserved

(c) The relation is in BCNF

(d) No null values exist

Correct Option: (a) No data is lost during decomposition

Justification: A lossless-join decomposition ensures that after decomposing a relation into smaller relations, joining those relations produces the original relation without losing information or generating incorrect tuples. It does not necessarily guarantee that all functional dependencies are preserved. It also does not automatically mean the relation is in BCNF or that null values are eliminated. Therefore, the primary purpose of lossless decomposition is to ensure no information is lost.